Review plan · for the teacher

How AI models work: unit check

AI · Level 3 · Unit check

Time	Materials
40 min	Board and chalk; The quiz sheet (one per team)

Learning goals

- One neuron
- How a network learns
- Train, test, trust?
- Inside a language model

Activity: Team quiz

1. Warm-up (5 min): Ask for one word that sums up each of the four lessons, and write them on the board.
2. Team quiz (15 min): In teams of four, read each question aloud; teams agree on one answer and write it on their sheet. Reveal the answer after each question, and ask one team to explain why.
3. Explain (8 min): Find the two questions most teams got wrong. Each team explains one of them on the board, using a drawing (a neuron, a hill, a test set or the loop).
4. Claim check (7 min): Read out one real AI claim (from an advert, an app or the news). Teams ask the three questions: tested on whom, on which data, with which mistakes? Agree what you would still need to know.
5. Wrap-up (5 min): Exit ticket: each person writes one idea from this unit they will use this week.

Print the quiz sheet (one per team) and read the other questions aloud from this check. Find one real AI claim to check.

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- Which AI claim in your school deserves the three questions?

Exit ticket

- One idea I will use, and when I will use it

Try it at school tomorrow
Choose one AI tool your school uses or is being offered, and ask its supplier the three questions: tested on whom, on which data, with which mistakes? Share the answers with a colleague.

Quiz sheet · for each team

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Team _____ Date _____

1 **A neuron has inputs 1 and 1, weights 2 and −3, and a bias of 2. What does it say?**

◯ Yes: the total is 7 ◯ No: the total is -3 ◯ No: the total is -1 ◯ Yes: the total is 1

2 **One neuron can learn to say yes only when the temperature is between 18 and 26 °C.**

◯ True ◯ False

3 **Put the training of a network in order.**

☐ Nudge every weight a little downhill

☐ Repeat with the next example

☐ Guess an answer for an example

☐ Start with random weights

☐ Measure the error against the label

4 **Match each word to its meaning.**

Weight	• • • •	The size of each training step
Learning rate	• • • •	How much an input matters
Test set	• • • •	How the newest token weighs earlier ones
Attention	• • • •	Examples the model never trains on

5 **During training, a model's error on its training examples keeps falling, but its error on new examples starts to rise. What is happening?**

◯ It is learning the real pattern ◯ The learning rate is too small ◯ It needs fewer examples ◯ It is overfitting

6 **Which of these are false negatives? Choose all that apply.**

☐ A spam filter lets a scam text into the inbox ☐ A cheating detector flags an honest essay

☐ A malaria test says "clear" for a child who has malaria ☐ A mobile-money app blocks a genuine payment as fraud

7 **Complete the sentence.**

A language model is _____ on huge amounts of text, then _____ with examples and people's ratings. It writes _____ text, so check the facts.

Word bank: fine-tuned · memorised · true · pretrained · likely

8 **Only 5 pupils in 100 need glasses. An eye-screening app that is 95% accurate must be finding most of them.**

◯ True ◯ False

9 **In the network playground, the error falls slowly but steadily, and is still falling when the playground stops at 3,000 rounds. What would you try?**

◯ Stop: it will never learn ◯ Fewer neurons ◯ A somewhat bigger learning rate ◯ Fewer training examples

10 **Which of these make a chatbot's answers more reliable for your class? Choose all that apply.**

☐ Give it the textbook page and ask it to answer only from that ☐ Use the first answer if it reads well

☐ Ask it to sound more confident ☐ Ask it whether it is sure ☐ Check its answer against the source

Answers · for the teacher

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1 **Yes: the total is 1**

1 × 2 + 1 × (−3) = −1, and the bias of 2 brings it to 1: above zero, so yes.

2 **False**

"Between 18 and 26" is yes in the middle and no on both sides.

3 **Top to bottom: 4, 5, 2, 1, 3**

Training starts from random weights; each round guesses, measures the error, and nudges the weights to shrink it, then moves on.

4 **Weight — How much an input matters**
Learning rate — The size of each training step
Test set — Examples the model never trains on
Attention — How the newest token weighs earlier ones

These four ideas run through the unit: weights hold what a network learns, the learning rate sets how fast it learns, the test set shows how it will really do, and attention lets a language model use context.

5 **It is overfitting**

Better on its own examples but worse on new ones is the signature of overfitting: it is fitting the noise.

6 **A malaria test says "clear" for a child who has malaria; A spam filter lets a scam text into the inbox**

In the first two, the tool said no (no malaria, not a scam) and the truth was yes: misses, so false negatives.

7 **A language model is pretrained on huge amounts of text, then fine-tuned with examples and people's ratings. It writes likely text, so check the facts.**

Pretraining comes first, on huge amounts of text; fine-tuning with examples and ratings follows.

8 **False**

An app that tells everyone "no glasses needed" is also 95% accurate and finds no one.

9 **A somewhat bigger learning rate**

A slow, steady fall means the steps go the right way but are small, so a somewhat bigger learning rate speeds things up; if the error starts jumping, step back down.

10 **Give it the textbook page and ask it to answer only from that; Check its answer against the source**

Grounding gives attention a trusted source to draw on, and checking catches what still slips through.